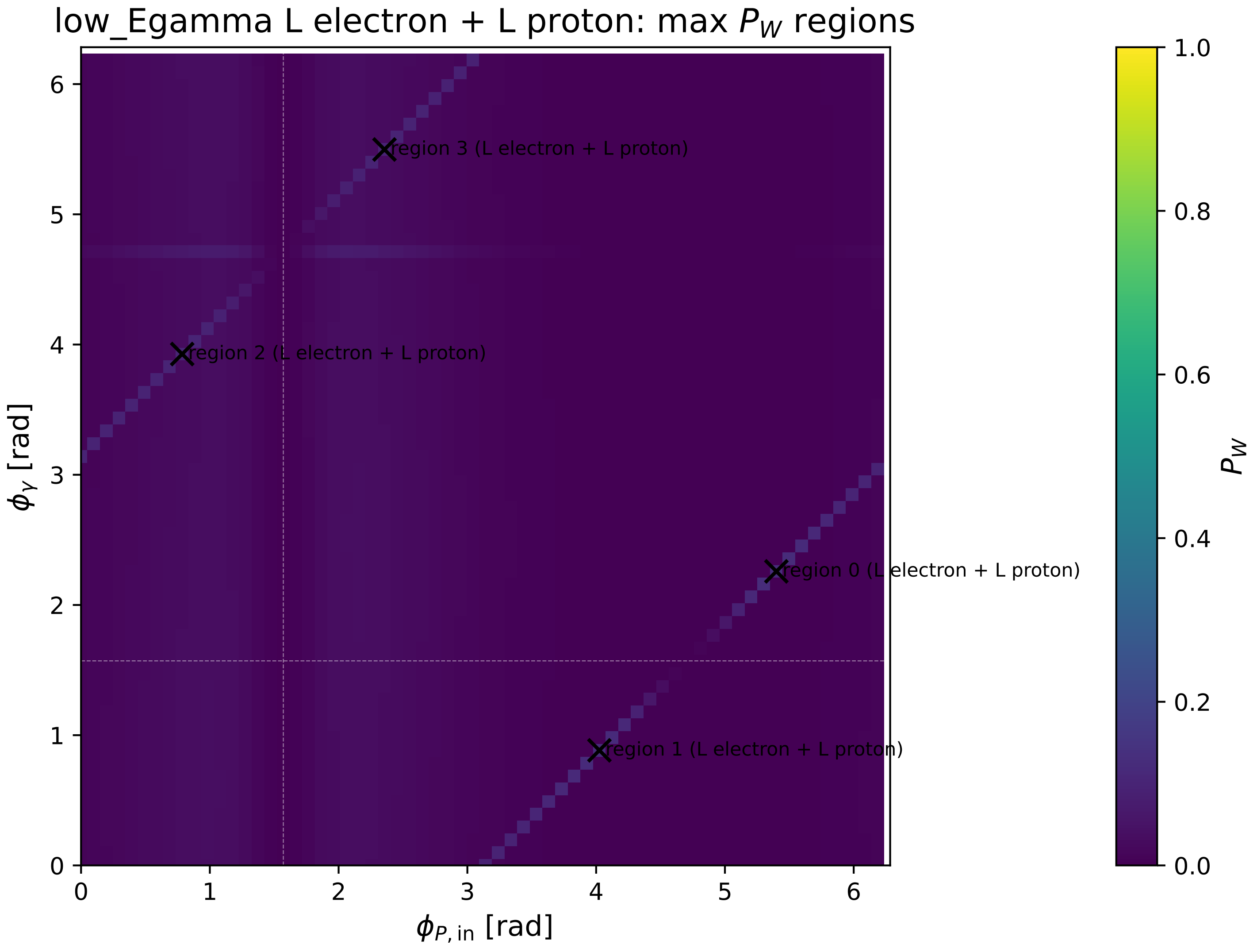
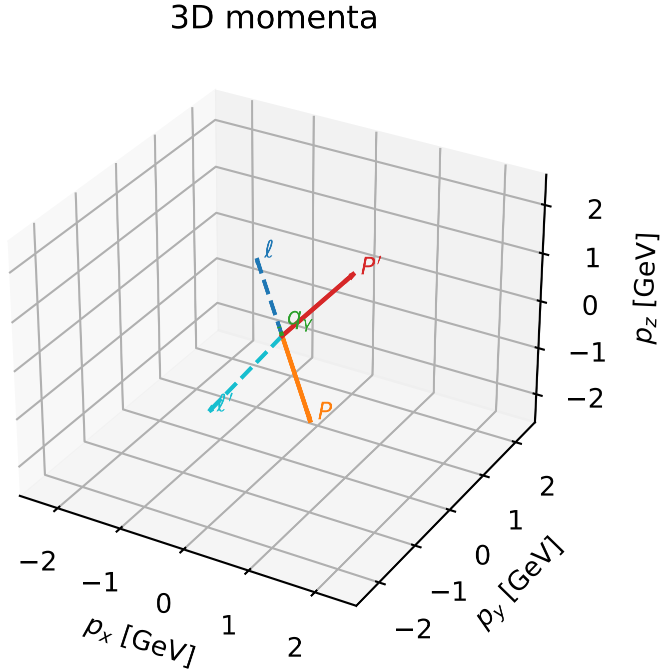

low_Egamma L electron + L proton: max P_W regions
region 3 (L electron + L proton)
region 2 (L electron + L proton)
region 0 (L electron + L proton)
region 1 (L electron + L proton)
ϕ_γ [rad]
$\phi_{P,\mathrm{in}}$ [rad]
P_W
0
1
2
3
4
5
6
0.0
0.2
0.4
0.6
0.8
1.0

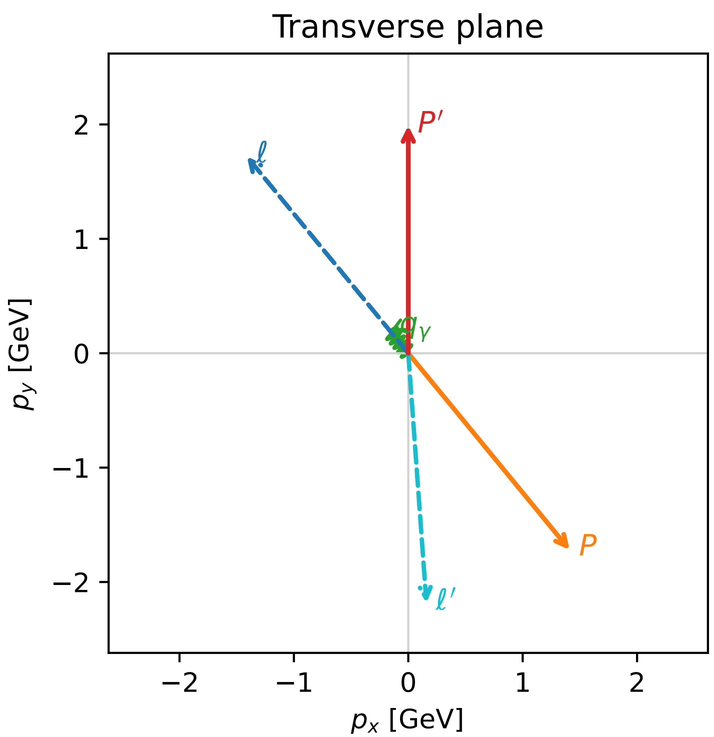
L electron + L proton characteristic max P_W configuration

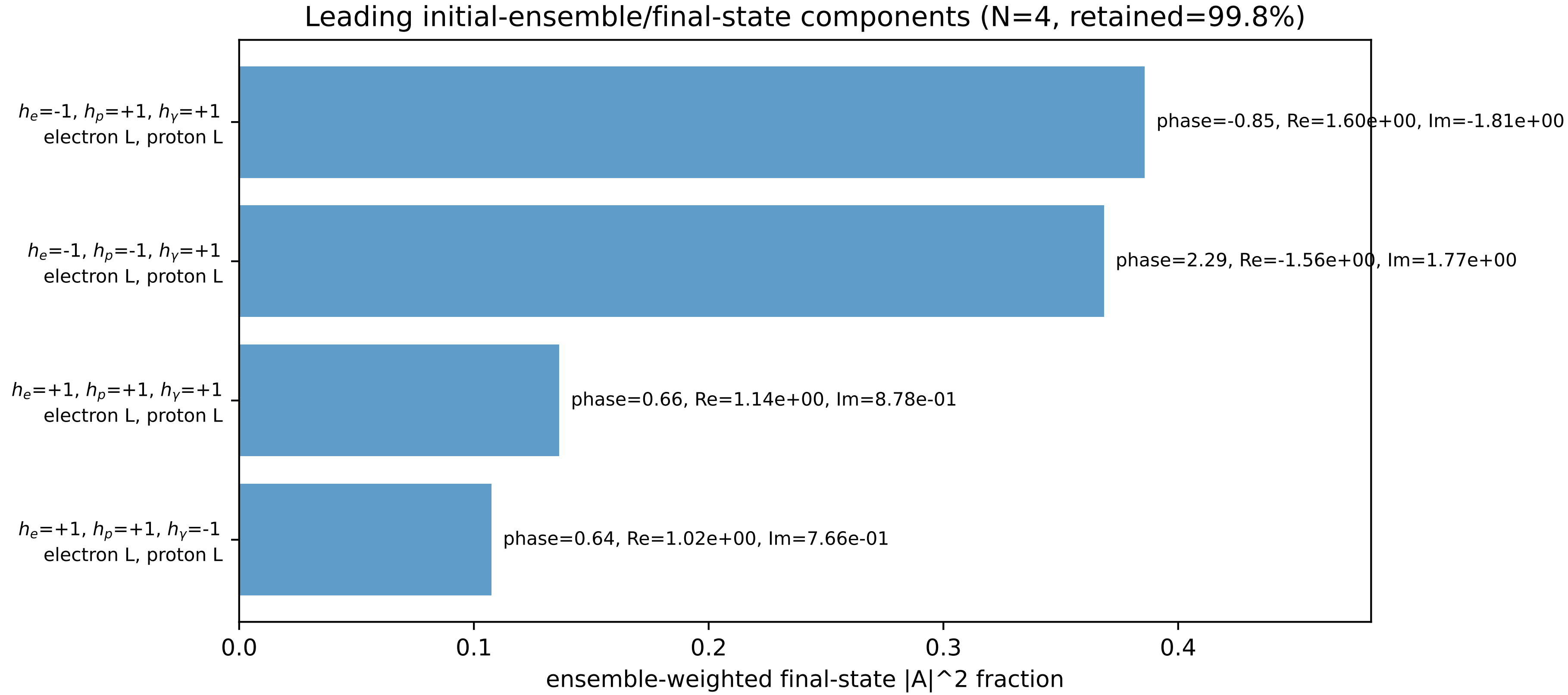


w_purity_low_egamma_ll_region_0 P_W=0.124066
region: low_Egamma_LL_region_0
outgoing-state purity: 1
kinematic point: high_s_high_theta_in_low_Egamma
incoming state: $|L_e\rangle \otimes |L_p\rangle$

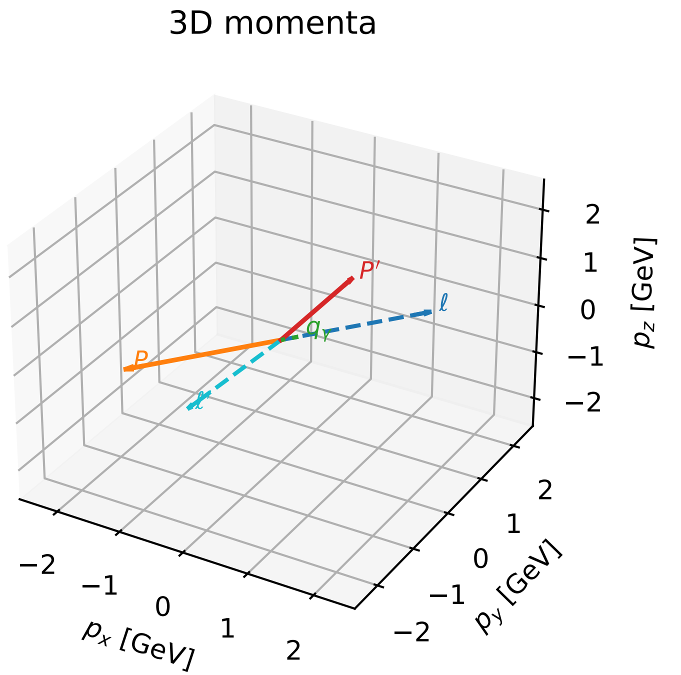
s=21.5158, $\sqrt{s}$=4.63852, $|\vec{P}|$=2.22442, $|\vec{P}'|$=1.98975
θ_{in}=1.57079, ϕ_ℓ=2.25802, ϕ_P=5.39961
E_γ=0.25, ϕ_γ=2.25802
Q^2=17.6924, x_B=0.981644, t=-15.7039, W^2=1.21068, y=0.873386
$\theta(\ell', \gamma)$=144.78 deg

four-momenta [E, p_x, p_y, p_z] GeV:
ℓ E= 2.2244 p_x= -1.4112 p_y= 1.7195 p_z= -0.0000
P E= 2.4141 p_x= 1.4112 p_y= -1.7195 p_z= 0.0000
ℓ' E= 2.1888 p_x= 0.1586 p_y= -2.1830 p_z= 0.0000
P' E= 2.1998 p_x= 0.0000 p_y= 1.9898 p_z= 0.0000
q_γ E= 0.2500 p_x= -0.1586 p_y= 0.1933 p_z= 0.0000





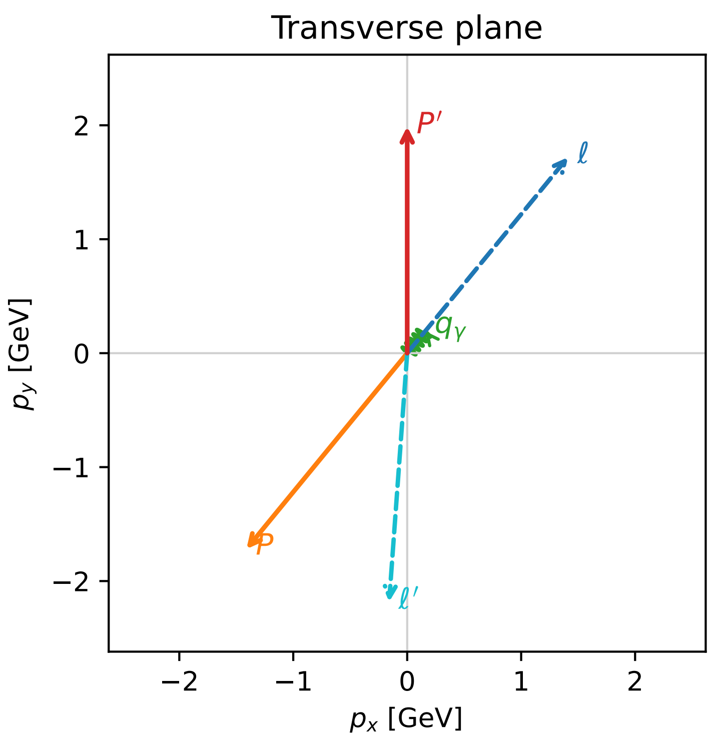
L electron + L proton characteristic max P_W configuration

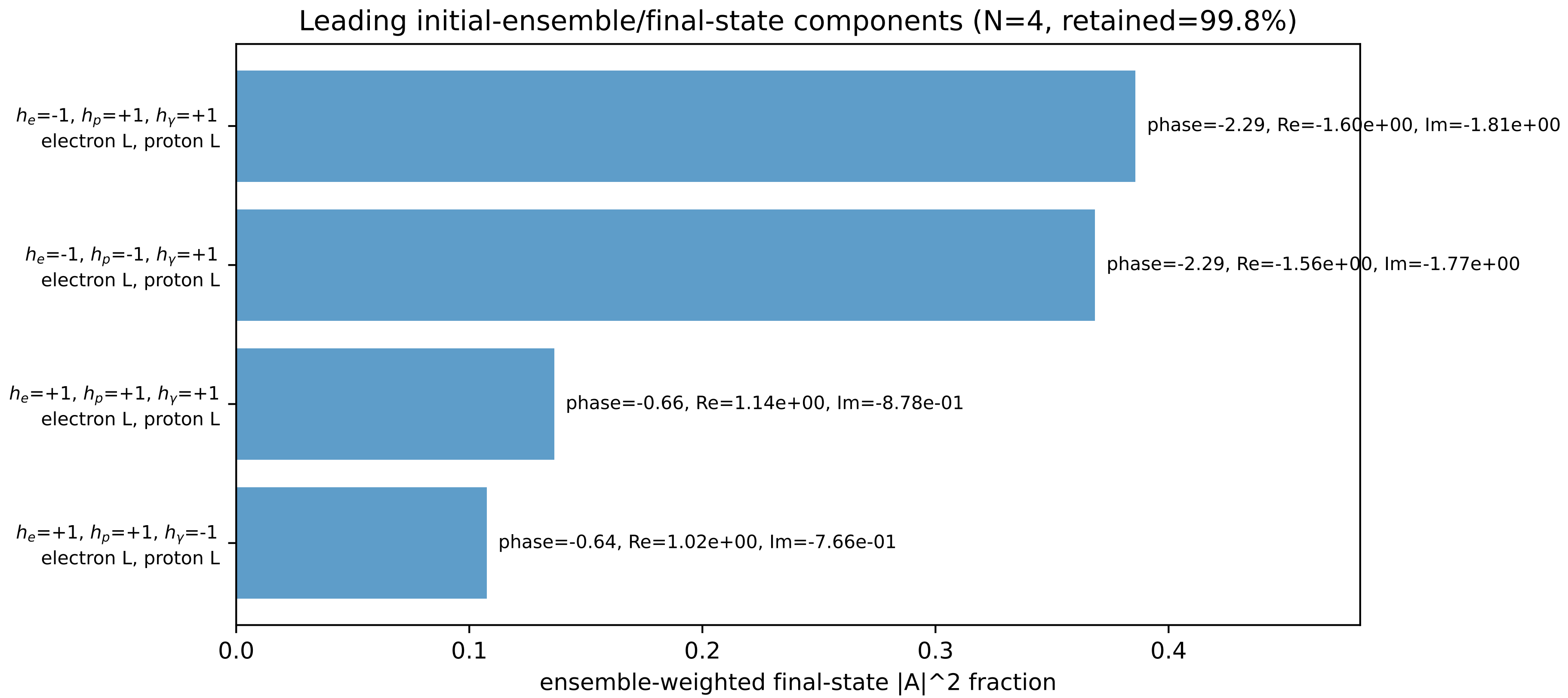


w_purity_low_egamma_ll_region_1 P_W=0.122127
region: low_Egamma_LL_region_1
outgoing-state purity: 1
kinematic point: high_s_high_theta_in_low_Egamma
incoming state: $|L_e\rangle \otimes |L_p\rangle$

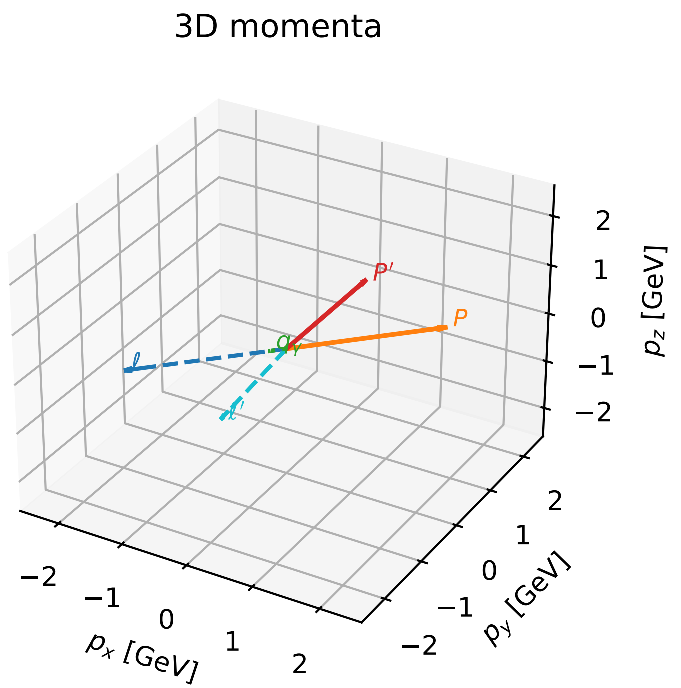
s=21.5158, $\sqrt{s}$=4.63852, $|\vec{P}|$=2.22442, $|\vec{P}'|$=1.98975
θ_{in}=1.57079, ϕ_ℓ=0.883573, ϕ_P=4.02517
E_γ=0.25, ϕ_γ=0.883573
Q^2=17.6924, x_B=0.981644, t=-15.7039, W^2=1.21068, y=0.873386
$\theta(\ell', \gamma)$=144.78 deg

four-momenta [E, p_x, p_y, p_z] GeV:
ℓ E= 2.2244 p_x= 1.4112 p_y= 1.7195 p_z= -0.0000
P E= 2.4141 p_x= -1.4112 p_y= -1.7195 p_z= 0.0000
ℓ' E= 2.1888 p_x= -0.1586 p_y= -2.1830 p_z= 0.0000
P' E= 2.1998 p_x= 0.0000 p_y= 1.9898 p_z= 0.0000
q_γ E= 0.2500 p_x= 0.1586 p_y= 0.1933 p_z= 0.0000





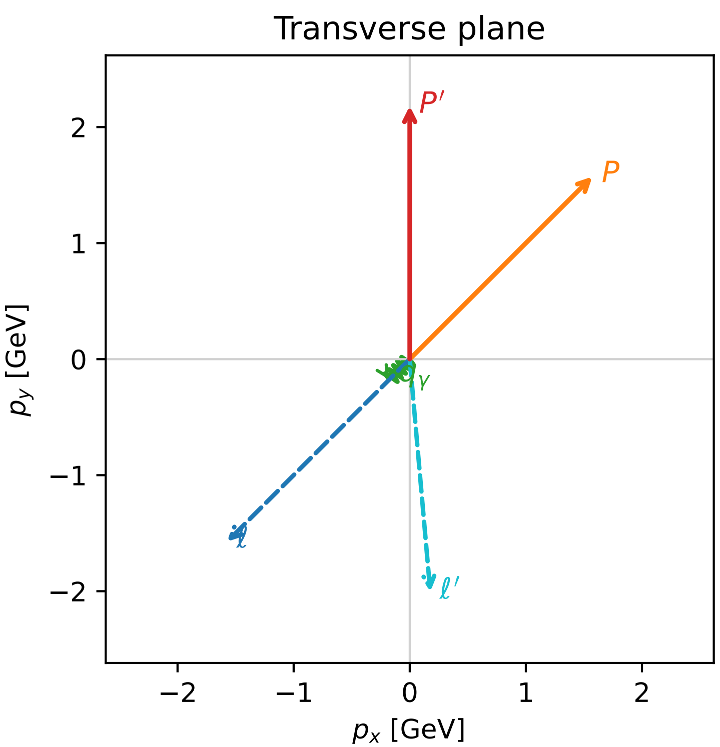
L electron + L proton characteristic max P_W configuration

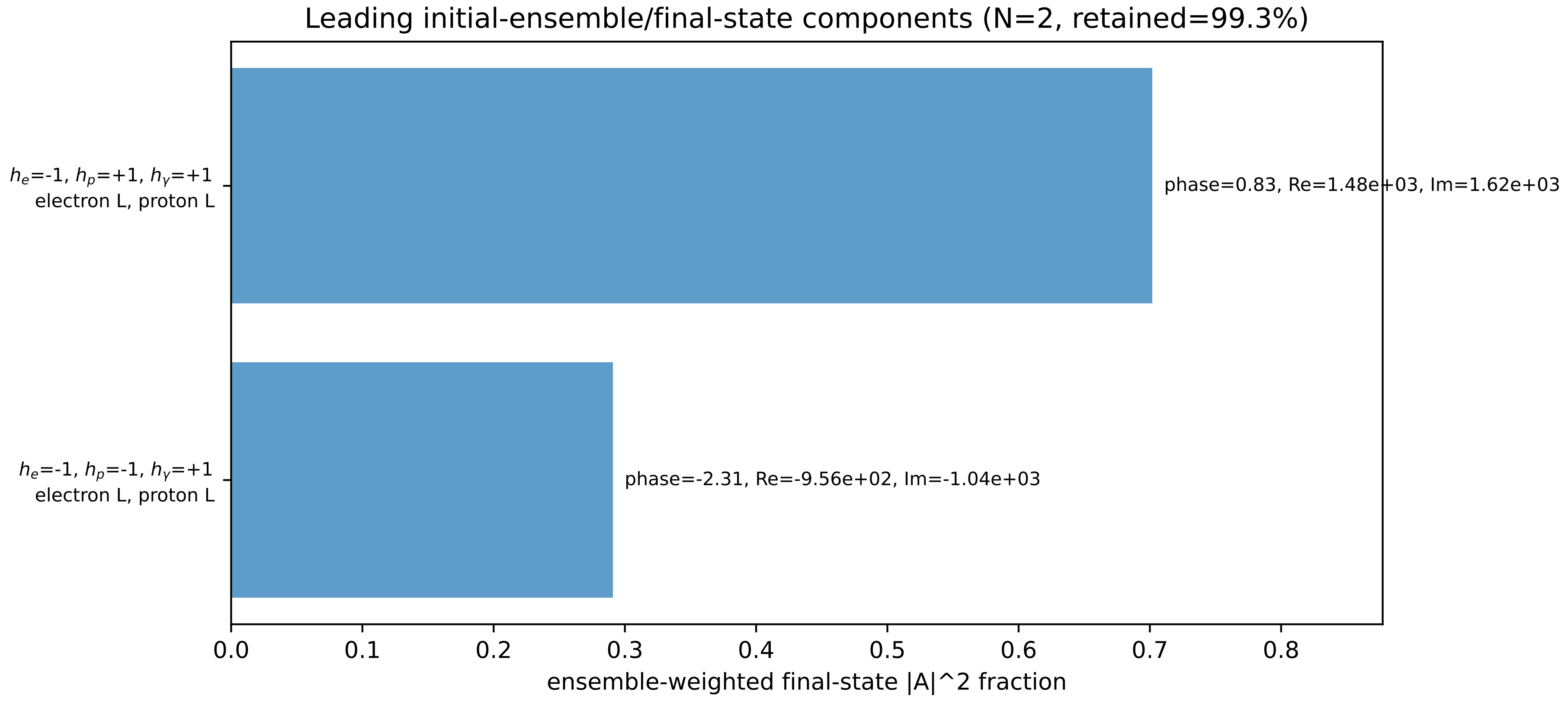


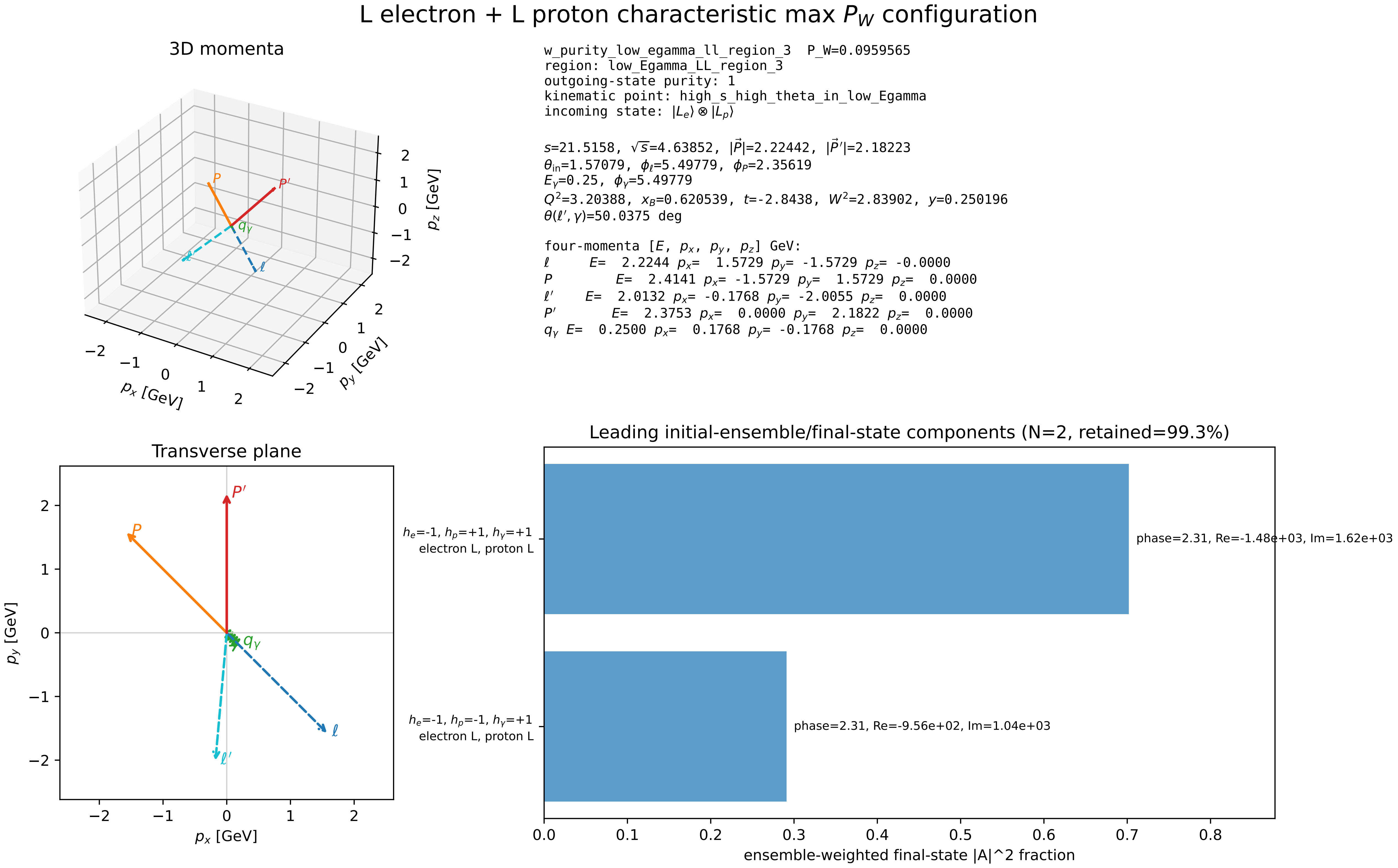
w_purity_low_egamma_ll_region_2 P_W=0.0983669
region: low_Egamma_LL_region_2
outgoing-state purity: 1
kinematic point: high_s_high_theta_in_low_Egamma
incoming state: $|L_e\rangle \otimes |L_p\rangle$

s=21.5158, $\sqrt{s}$=4.63852, $|\vec{P}|$=2.22442, $|\vec{P}'|$=2.18223
$\theta_{\rm in}$=1.57079, ϕ_ℓ=3.92699, ϕ_P=0.785398
E_γ=0.25, ϕ_γ=3.92699
Q^2=3.20388, x_B=0.620539, t=-2.8438, W^2=2.83902, y=0.250196
$\theta(\ell', \gamma)$=50.0375 deg

four-momenta [E, p_x, p_y, p_z] GeV:
ℓ E= 2.2244 p_x= -1.5729 p_y= -1.5729 p_z= -0.0000
P E= 2.4141 p_x= 1.5729 p_y= 1.5729 p_z= 0.0000
ℓ' E= 2.0132 p_x= 0.1768 p_y= -2.0055 p_z= 0.0000
P' E= 2.3753 p_x= 0.0000 p_y= 2.1822 p_z= 0.0000
q_γ E= 0.2500 p_x= -0.1768 p_y= -0.1768 p_z= 0.0000






L electron + L proton characteristic max P_W configuration
3D momenta
w_purity_low_egamma_ll_region_3 P_W=0.0959565
region: low_Egamma_LL_region_3
outgoing-state purity: 1
kinematic point: high_s_high_theta_in_low_Egamma
incoming state: $|L_e\rangle \otimes |L_p\rangle$
s=21.5158, $\sqrt{s}$=4.63852, $|\vec{P}|$=2.22442, $|\vec{P}'|$=2.18223
θ_{in}=1.57079, ϕ_ℓ=5.49779, ϕ_P=2.35619
E_γ=0.25, ϕ_γ=5.49779
Q^2=3.20388, x_B=0.620539, t=-2.8438, W^2=2.83902, y=0.250196
$\theta(\ell', \gamma)$=50.0375 deg
four-momenta [E, p_x, p_y, p_z] GeV:
ℓ E= 2.2244 p_x= 1.5729 p_y= -1.5729 p_z= -0.0000
P E= 2.4141 p_x= -1.5729 p_y= 1.5729 p_z= 0.0000
ℓ' E= 2.0132 p_x= -0.1768 p_y= -2.0055 p_z= 0.0000
P' E= 2.3753 p_x= 0.0000 p_y= 2.1822 p_z= 0.0000
q_γ E= 0.2500 p_x= 0.1768 p_y= -0.1768 p_z= 0.0000
p_x [GeV]
p_y [GeV]
p_z [GeV]
P
P'
q_γ
ℓ'
ℓ
Transverse plane
Leading initial-ensemble/final-state components (N=2, retained=99.3%)
h_e=-1, h_p=+1, h_γ=+1
electron L, proton L
phase=2.31, Re=-1.48e+03, Im=1.62e+03
h_e=-1, h_p=-1, h_γ=+1
electron L, proton L
phase=2.31, Re=-9.56e+02, Im=1.04e+03
ensemble-weighted final-state |A|^2 fraction